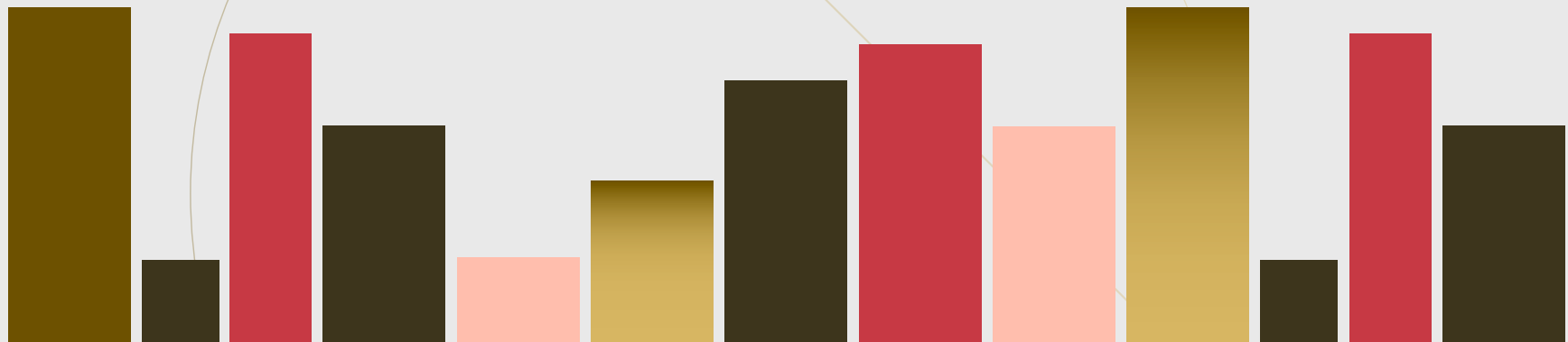


ZEN

IUT-1 - 9th
exercises.



25zs 04111

Stack @ Luna

1. (a) $e \in H, N$ then $e \cdot e = e \in HN$

If $h_1 n_1, h_2 n_2 \in HN$,

N is normal, so $\exists n'_1 \in N, h_2 n'_1 h_2^{-1} = n_1$

$$\underline{h_1 n_1} \cdot h_2 n_2 = h_1 \underline{h_2 n'_1 h_2^{-1}} \underline{h_2} h_2 n_2$$

$$= h_1 h_2 \overset{\substack{\uparrow \\ e}}{n'_1} n_2 \in \underline{HN}$$

If $hn \in HN, \exists n \in N, h^{-1}nh = n$

$$(hn)^{-1} = h^{-1}n^{-1}$$

$$= h^{-1}n^{-1}h h$$

$$= h^{-1}n^{-1} \in \underline{HN}$$

Q

(b) $e \in H, N$ then $e \in H \cap N$

If $a, b \in H \cap N$ $a, b \in H, N$ then $ab \in H, N$ then
 $ab \in H \cap N$

If $a \in H \cap N$ $a \in H, N$ then $a^{-1} \in H, N$ then $a^{-1} \in H \cap N$.

$H \cap N \subseteq H$ is subgroup.

Consider $h \in H$, $a \in H \cap N$.

N is normal, $\exists a'$, $h a' h^{-1} = a$.

$h^{-1} a' h \in H$ thus $a' \in H$ also $a' \in H \cap N$

$$\underline{ha} = h h^{-1} a' h = \underline{a' h} \quad \text{Therefore } H \cap N \text{ is normal!}$$

(c) f is well-defined

Take $h, g \in H$, assume $h(H \cap N) = g(N \cap H)$

$$\exists \underline{a \in N \cap H}, h = ga \text{ then } hN = gaN = gN$$


 $a \in N$

Thus, they jump into the same value,

f is injection

$$\text{Assume } f(h(H \cap N)) = f(g(N \cap N))$$

$$\Rightarrow hN = gN$$

$$H \ni g^{-1}h = n$$

$$\Leftrightarrow \exists n \in N. \quad h = gn \Rightarrow n \in H \quad \text{thus } n \in H \cap N$$

$$\underline{h(H \cap N)} = \underline{gN(H \cap N)} = \underline{g(H \cap N)}$$

f is surjection

For any $h \in H$, we can take $h(N \cap H)$ which's mapped into hN

f is isomorphic

$$e) \quad \underline{f(e(N \cap H)) = eN}$$

$$ab) \quad \underline{f(\underline{a(N \cap H) \cdot b(N \cap H)})} = \underline{f(\underline{ab(N \cap H)})}$$

$$= abN = \underline{aN} \cdot \underline{bN}$$

$$= \underline{f(a(N \cap H))} \cdot \underline{f(b(N \cap H))}$$

$$a^{-1}) \quad \underline{f(a^{-1}(N \cap H))}$$

$$= a^{-1}N$$

$$= (aN)^{-1}$$

$$= \underline{f(a(N \cap H))^{-1}}$$



2. (a) $H \trianglelefteq G$, $K \trianglelefteq G$, by 1-(b), $K \trianglelefteq G \cap H = H$

$$\forall g \in G$$

$$h \in H$$

$$\exists h' \in H$$

$$\underline{gK \cdot hK = ghK}$$

$$= h'gK = \underline{h'K \cdot gK}$$

$$\text{Thus, } gK \cdot (H/K) \cdot (gK)^{-1} = H/K \quad \blacksquare$$

2. (b)

f is well-defined.

$$\text{If, } g_1 = g_2 k, \quad g_1 H = g_2 k H = g_2 H \quad (\text{since } k \in K \trianglelefteq H)$$

f is surjective.

For any $gH \leftarrow gK$

f is homomorphic

$$e) \quad f(ek) = eH$$

$$ab) \quad f(ak \cdot bk) = f(abk) = abH = aH \cdot bH = f(ak) \cdot f(bk)$$

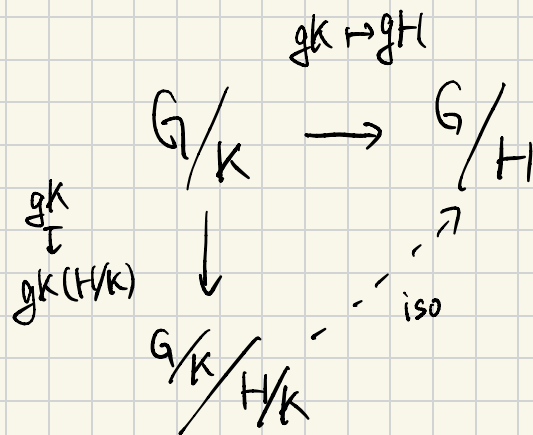
$$a') \quad f(a^{-1}k) \cdot a^{-1}H = (aH)^{-1} = f(ak)^{-1}$$

Kernel

$$\text{If } f(aK) = eH,$$

which means $a \in H$.

Such aK is in H/K ■



3. (a) By 15.3 discussion,

$$|Gx| = |G : G_x| \quad |G| = p^n$$

which means $|Gx| = 1$ or p or $\dots$ or p^n

Here, $|Gx| = 1 \iff Gx = \{x\} \iff x \text{ is fixed elem.}$

$$|X| = \sum |Gx|$$

$$= \sum_{|Gx|=1} |Gx| + \sum_{|Gx| \neq 1} |Gx|$$

$$\equiv \sum_{|Gx|=1} |Gx| = \text{no. of fixed points. (mod } p) \quad \blacksquare$$

(b) If $|X| \not\equiv 0 \pmod p$, at least one x exists s.t. $|Gx| = 1$, which's fixed element. $\blacksquare$

4. (a) $ege^{-1} = g$, so $e \in Z(G) \Rightarrow |Z(G)| \geq 1$.

$$|G : N_G(x)| \mid |G| = p^n$$

$$\Rightarrow |G : N_G(x)| = 1 \text{ or } p \text{ or } \dots \text{ or } p^n$$

Here, $|G : N_G(x)| = 1 \Leftrightarrow N_G(x) = G \Leftrightarrow x \in Z_G(G)$

$$\text{So, } \underline{p^n} = |G| = |Z(G)| + \sum_{|G : N_G(x)| > 1} \underline{|G : N_G(x)|} \equiv 0 \pmod p$$

$$\underline{0} \equiv |Z(G)| \pmod p$$

$\not\equiv 1$, thus $Z(G) \neq \{e\}$ $\blacksquare$

(b) For odd primes, by Feit-Thompson theorem.

Consider $p=2$.

For $n=1$, $G \cong \{1, a\}$ with $a \cdot a = 1$, which is solvable.

As induction, assume for $k \leq k$, 2^l order groups are solvable.

If $|G| = 2^{k+1}$, $Z(G) \geq 2$

$$2^{k+1} = |G| = |G/Z(G)| \cdot |Z(G)|$$

so $|G/Z(G)| < 2^l$ for some $l \leq k$,

which means $G/Z(G)$ is solvable,

$Z(G)$ is abelian thus solvable.

Then G is also solvable. $\blacksquare$

5. Take natural homomorphism $\pi : G \rightarrow G/[G, G]$

G_m is subgroup.

$\forall x, y \in G_m$

$$\begin{aligned}\pi((xy)^m) &= \pi(xy)^m = (\pi(x)\pi(y))^m \\ &= \pi(x)^m \pi(y)^m \\ &= \pi(x^m) \pi(y^m) \\ &= e\end{aligned}$$

$G/[G, G]$ is abelian.

thus, $(xy)^m \in [G, G]$

And $xg \in G_m$.

G_m is normal

Assume $\forall g \in G, x \in G_m$

$$(gxg^{-1})^m = gx^m g^{-1}$$

$[G, G]$ is also normal, so $gx^m g^{-1} \in [G, G]$
" $(gxg^{-1})^m$

$$\Rightarrow g x g^{-1} \in G_m. \quad \square$$